

# Linear Algebra: Guided Review and Exercises

**Purpose.** This sheet is meant to be a learning document. Some of the computational procedures below were not developed in detail in lecture. Each such topic therefore begins with a short introduction and a worked example, followed by exercises.

Unless otherwise stated, all vector spaces are finite-dimensional over  $\mathbb{R}$  or  $\mathbb{C}$ .

## I. Solving Linear Systems by Elimination

**Mini-lecture.** A linear system is best viewed through its *augmented matrix*. The three elementary row operations are:

- (i) interchange two rows;
- (ii) multiply a row by a nonzero scalar;
- (iii) add a multiple of one row to another row.

These operations do not change the solution set. The practical goal is echelon form: identify pivot variables, free variables, and any contradictory row of the form

$$(0 \ \cdots \ 0 \mid c), \quad c \neq 0.$$

A contradictory row means no solution. If every variable is a pivot variable, the solution is unique. If the system is consistent and at least one variable is free, there are infinitely many solutions.

**Worked example.** Consider

$$\begin{aligned}x + y + z &= 6, \\2x + y + z &= 8, \\x + 2y + z &= 9.\end{aligned}$$

Start from

$$\left( \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 8 \\ 1 & 2 & 1 & 9 \end{array} \right).$$

Use  $R_2 \leftarrow R_2 - 2R_1$  and  $R_3 \leftarrow R_3 - R_1$ :

$$\left( \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -4 \\ 0 & 1 & 0 & 3 \end{array} \right).$$

Swap the last two rows and then add the new second row to the third:

$$\left( \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -1 \end{array} \right).$$

Thus  $z = 1$ ,  $y = 3$ , and  $x = 2$ . The point of the procedure is not the arithmetic itself: the pivot structure tells us whether a solution exists and how many degrees of freedom it has.

**1. Practice.** For each system below:

(a) write the augmented matrix;

(b) row-reduce to echelon form;

(c) identify pivot and free variables;

(d) state whether the system has a unique solution, no solution, or infinitely many solutions;

(e) if there are infinitely many solutions, parameterize the complete solution set.

(i)

$$2x + 2y = 5,$$

$$x - 4y = 0;$$

(ii)

$$x - 3y + z = 1,$$

$$x + y + 2z = 14;$$

(iii)

$$-x - y = 1,$$

$$-3x - 3y = 4.$$

For every consistent nonunique system, write the solution in the form

$$x = x_p + x_h,$$

where  $x_p$  is one particular solution and  $x_h$  ranges over the null space of the coefficient matrix.

i. a)  $\begin{bmatrix} 2 & 2 & | & 5 \\ 1 & -4 & | & 0 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_1} \begin{bmatrix} 2 & 2 & | & 5 \\ 0 & -5 & | & -2\frac{1}{2} \end{bmatrix}$

b)  $\begin{bmatrix} 2 & 2 & | & 5 \\ 1 & -4 & | & 0 \end{bmatrix}$

c)  $x, y$  pivots

d) unique,  $y = 1/2$   $x = 2$

$$\Rightarrow \vec{x} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

e) N/A

$$\begin{aligned}x - 3y + z &= 1, \\x + y + 2z &= 14;\end{aligned}$$

$$i), a, b) \begin{bmatrix} 1 & -3 & 1 & | & 1 \\ 1 & 1 & 2 & | & 14 \end{bmatrix} \xrightarrow{-R_1} \begin{bmatrix} 1 & -3 & 1 & | & 1 \\ 0 & 4 & 1 & | & 13 \end{bmatrix}$$

c)  $x, y$  pivot,  $z$  free

d) infinite solutions

e) let  $z = t$

$$\Rightarrow y = 3t \Rightarrow \vec{x} = \begin{bmatrix} 4t \\ 3t \\ t \end{bmatrix}$$

$$\text{All sol: } t \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} \text{ for } t \in \mathbb{R}, \vec{x}_1 = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

$$\begin{aligned}-x - y &= 1, \\-3x - 3y &= 4.\end{aligned}$$

$$iii), a, b) \begin{bmatrix} -1 & -1 & | & 1 \\ -3 & -3 & | & 4 \end{bmatrix} \xrightarrow{-3R_1} \begin{bmatrix} -1 & -1 & | & 1 \\ 0 & 0 & | & 1 \end{bmatrix}$$

c) 2 pivot, 1 free

d) no solution, inconsistent

$$\text{row } 0x + 0y = 1$$

e) N/A

2. A parameter and consistency. Consider

$$3x + 2y = 10,$$

$$6x + 4y = b.$$

$$A = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

- (a) Row-reduce the augmented matrix without assigning a value to  $b$  at the outset.
- (b) For which values of  $b$  is the system consistent?
- (c) When it is consistent, parameterize all solutions.
- (d) Compare the rank of the coefficient matrix with the rank of the augmented matrix in the consistent and inconsistent cases.

$$a) A^* = \begin{bmatrix} 3 & 2 & | & 10 \\ 6 & 4 & | & b \end{bmatrix} \xrightarrow{-2R_1} \begin{bmatrix} 3 & 2 & | & 10 \\ 0 & 0 & | & b-20 \end{bmatrix} = R$$

$$b) b = 20 \text{ only}$$

$$c) \text{ Let } b = 20 \Rightarrow \begin{bmatrix} 3 & 2 & | & 10 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$3x + 2y = 10$$

Let  $y = t$  be free variable,

$$x = \frac{10 - 2t}{3} = \frac{10}{3} - \frac{2}{3}t$$

$$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{10}{3} - \frac{2}{3}t \\ 0 + 1t \end{bmatrix} = \begin{bmatrix} 10/3 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2/3 \\ 1 \end{bmatrix}$$

$$\vec{x} = \vec{x}_0 + t \vec{v}$$

consistent  $b = 20$

inconsistent  $b \neq 20$

$$d) \rho(A) = 2, \rho(A^*) = 1; \rho(A) = 2, \rho(A^*) = 2$$



## II. Rank, Column Space, and Null Space

**Mini-lecture.** Row reduction reveals rank and the null space. If the reduced matrix has  $r$  pivot columns, then the matrix has rank  $r$ . To find a basis of the *column space*, use the corresponding pivot columns of the *original* matrix. To find the null space, solve  $Ax = 0$  using the reduced system and parameterize by the free variables.

**Worked example.** Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}.$$

Row reduction gives

$$A \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence  $\text{rank } A = 2$ . The first two columns are pivot columns, so a basis for the column space is formed by the first two columns of the *original* matrix:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

For  $Ax = 0$ , write  $x_3 = t$ . Then  $x_1 = t$  and  $x_2 = -2t$ , so

$$\text{Ker } A = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

Notice the rank-nullity check:  $2 + 1 = 3$ , the number of columns of  $A$ .

**3. Practice.** For each matrix below, use row reduction to find:

- (a) its rank;
- (b) a basis for its column space;
- (c) a basis for its null space;
- (d) the dimensions predicted by rank-nullity.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{pmatrix}.$$

Finally, explain how your calculations determine whether the columns of each matrix are linearly independent.

A. a)  $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow{-3R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow{-2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & -1 & -5 \end{pmatrix}$   
 $\xrightarrow{2R_3} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -5 & -7 \end{pmatrix} \xrightarrow{-5R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{pmatrix}$

b) all pivots, all original cols.  
 $AB = \left\{ \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}$

c) None (null space = 0)

d)  $\dim = 3$

$$B. a) \begin{pmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{pmatrix} \xrightarrow{+3R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 2 & -3 & 1 \end{pmatrix}$$

$$\xrightarrow{-2R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & -7 & 7 \end{pmatrix} \xrightarrow{+R_2, R_2/7} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

b) 2 pivots, col 1 & 2

$$B_B = \left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} \right\}$$

$$c) B_{\vec{x}} = \vec{0}$$

$$x + 2y - 3z = 0$$

$$-3x + y + 2z = 0$$

$$2x - 3y + z = 0$$

$$\text{let } z = t \quad x = t, y = t$$

$$\text{Ker } B = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$d) \dim B = 3 = \dim \text{Col } B + \dim \text{Ker } B$$

4. **Proof: rank-nullity without coordinates.** Suppose  $T : \mathbb{F}^4 \rightarrow \mathbb{F}^2$  is linear and

$$\text{Ker } T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_1 = 5x_2, x_3 = 7x_4\}.$$

Without constructing a matrix for  $T$ , prove that  $T$  is surjective.

Claim:  $T$  is surjective i.f.f

$\text{Im } T = \mathbb{F}^2$ , the entire codomain

By rank nullity then,

$$\dim \mathbb{F}^4 = \dim \text{Im } T + \dim \text{Ker } T$$

Since  $\text{Ker } T = \{(5x_2, x_2, 7x_4, x_4)\}$

$T$  maps all  $x_2$  &  $x_4$   
to the zero vector in  $\mathbb{F}^2$ ,

$$\text{s.t. } \dim \text{Ker } T = 2$$

$$\dim \mathbb{F}^4 = 4 = \dim \text{Im } T + 2$$

$$\text{s.t. } \dim \text{Im } T = 2$$

since  $\dim \mathbb{F}^2 = 2$ ,  $\text{Im } T$  is

the entire codomain, thus surj.  $\square$

5. **Proof: spanning and independence as properties of a map.** Let  $v_1, \dots, v_m \in V$ , and define

$$T: \mathbb{F}^m \rightarrow V, \quad T(z_1, \dots, z_m) = z_1 v_1 + \dots + z_m v_m.$$

Prove directly from the definitions that

(a)  $v_1, \dots, v_m$  spans  $V$  if and only if  $T$  is surjective;

(b)  $v_1, \dots, v_m$  is linearly independent if and only if  $T$  is injective.

$$a) \Rightarrow \text{if } \text{span}(v_1, \dots, v_m) = V, \text{ then}$$

$$\forall v \in V, \forall z_i \in \mathbb{F}, v = z_1 v_1 + \dots + z_m v_m$$

$$\text{Since } T(z_1, \dots, z_m) = \sum_{i=1}^m z_i v_i,$$

$$\text{span}(v_i) = V = z_1 v_1 + \dots + z_m v_m = T(z_i)$$

$$T \text{ surj. iff } \text{Im } T = V$$

$$\text{Im } T = T(z_1, \dots, z_m) = V$$

$$\Rightarrow T \text{ surj.}$$

$$\Leftarrow \text{if } T \text{ surj, } \text{Im } T = V$$

$$\text{s.t. } T(z_1, \dots, z_m) = V$$

$$z_1 v_1 + \dots + z_m v_m = v \text{ for all } v \in V$$

$$V = \text{Im } T = T(z_1, \dots, z_m) = \text{span}(v_1, \dots, v_m)$$

$$+ \text{W.S. } V = \text{span}(v_1, \dots, v_m)$$

$\Rightarrow$  if  $v_1, \dots, v_m$  lin ind.,  
 the only  $z_i \in \mathbb{F}$  that solve  
 $z_1 v_1 + \dots + z_m v_m = 0$  are  $z_1 = \dots = z_m = 0$   
 therefore the only lin comb  $\sum_{i=1}^m z_i v_i = 0$   
 is  $\vec{z} = (0, \dots, 0) = \vec{0}_{\mathbb{F}}$

since  $T(z_1, \dots, z_m) = z_1 v_1 + \dots + z_m v_m$   
 the only  $T(\vec{z}) = 0_v$  is  $\vec{0}_{\mathbb{F}}$

$$\Rightarrow \ker T = \vec{0}_{\mathbb{F}}$$

$T$  is inj. iff  $\ker T = \{\vec{0}_{\mathbb{F}}\}$ ,

$\Rightarrow T$  is inj.

$\Leftarrow$  if  $T$  inj,  $\ker T = \{\vec{0}_{\mathbb{F}}\}$

$$\text{s.t. } T(\vec{z}) = 0 = z_1 v_1 + \dots + z_m v_m$$

only for  $z_1 = \dots = z_m = 0$ ,

$$\text{since } \vec{0}_{\mathbb{F}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix},$$

thus the only  $\vec{z}$  that solves  $\sum_{i=1}^m z_i v_i = 0$  is  $z_1 = \dots = z_m = 0$ , thus lin ind.

6. **Proof: rank one means an outer product.** Let  $A$  be a nonzero  $m \times n$  matrix. Prove that  $\text{rank } A = 1$  if and only if there exist vectors  $c \in \mathbb{F}^m$  and  $d \in \mathbb{F}^n$  such that

$$A = cd^T.$$

Unsure

Equivalently, prove that  $A_{jk} = c_j d_k$  for all  $j, k$ .

( $\Rightarrow$ ) if  $\text{rank } A = 1$ ,  
 either  $m = n = 1$ , s.t.  $A = a$   
 then  $\exists c, d \in \mathbb{F}$  s.t.  $a = cd$   
 for all  $a \in \mathbb{F}$  by  
 closure of a field

or  $m > 1, n > 1$ ,

$$\text{s.t. } A^{m \times n} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

since  $\text{rank } A = 1$ ,  $\text{col } A$  is gen,  
 $\dim \text{Col } A = 1$  s.t.

$\text{Col } A = \text{span} \{a_1, \dots, a_n\}$   
 is 1 dim subspace of  $\mathbb{F}^m$   
 ie a line in  $\mathbb{F}^m$

thus  $\exists c \in \mathbb{F}^m : c \neq 0$  s.t.

$\{c\}$  is basis for  $\text{col } A$

$\Rightarrow \exists d_i \in \mathbb{F}$  s.t.  $a_i = d_i c$

let  $\vec{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}$  where  $d_i \in F$

$$\Rightarrow A = \begin{bmatrix} a_1 & \dots & a_n \\ 1 & & 1 \end{bmatrix} = \begin{bmatrix} d_1 & \dots & d_n \\ 1 & & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} c \\ \vdots \\ c \end{bmatrix} [d_1 \dots d_n] = \vec{c} \vec{d}^T$$

thus  $A_{jk} = (\vec{c} \vec{d}^T)_{jk} = c_j d_k$

$\Leftarrow$  if  $A = \vec{c} \vec{d}^T$ ,  $c \neq 0, d \neq 0$   
since  $A \neq 0$

thus for at least one

$$c_j \in \vec{c}, d_k \in \vec{d},$$

it must hold  $c_j \neq 0, d_k \neq 0$

have  $A_{jk} = c_j d_k \neq 0$

thus  $\text{rank}(A) \geq 1$



for any  $x \in \mathbb{F}^n$ ,

$$Ax = (cd^T)x = c(d^Tx) = (d^Tx)c$$

since  $d^Tx$  is scalar in  $\mathbb{F}$ ,

$$\text{im } A \subseteq \text{span}\{c\}$$

since  $c \neq 0$ ,

$$\dim(\text{span}\{c\}) = 1,$$

thus  $\dim(\text{im } A) \leq \dim(\text{span}\{c\}) = 1$

s.t.  $\dim(\text{im } A) \leq 1$

since  $A \neq 0$ ,  $\dim(\text{im } A) > 0$

thus  $\dim(\text{im } A) = 1$

$\text{rank } A = 1$

□

### III. Linear Maps, Invertibility, and Direct Sums

**Checkpoint.** The next problems are deliberately proof-oriented. They connect the calculations above to the basis-free statements used in lecture.

7. Let  $A = (A_{jk})$  be an  $n \times n$  matrix. Prove that the following are equivalent:

(a) the homogeneous system  $Ax = 0$  has only the zero solution;

(b) for every  $b \in \mathbb{F}^n$ , the system  $Ax = b$  has a solution.

Explain why this is the matrix form of the equivalence between injectivity and surjectivity for a linear operator on a finite-dimensional vector space.

$\Rightarrow$  if a),  $Ax=0$  has only  
 $\vec{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$  solution  
thws for  $A = \begin{bmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{bmatrix}$ ,  
 $\{a_1, \dots, a_n\}$  is lin ind set  
s.t.  $x_1 \vec{a}_1 + x_2 \vec{a}_2 + \dots + x_n \vec{a}_n = \vec{0}$   
only for  $x_1 = \dots = x_n = 0$   
thws  $\text{col } A = \text{span } \{a_1, \dots, a_n\}$   
 $\text{im } A = \mathbb{F}^n$   
 $\Rightarrow \forall \vec{b} \in \mathbb{F}^n, \exists \vec{x} \in \mathbb{F}^n$  s.t.  
 $x_1 \vec{a}_1 + \dots + x_n \vec{a}_n = \vec{b}$   
thws  $Ax=b$  has solution  
for any  $b \in \mathbb{F}^n$

$\Leftarrow$  if  $\forall b \in \mathbb{F}^n, \exists x \in \mathbb{F}^n$   
s.t.  $Ax = b$ ,

for a given  $x$ ,

$Ax$  is unique

thus for  $b = 0$ , there

exists  $x_0 \in \mathbb{F}^n$  s.t.  $Ax = b = 0$

since  $\text{im } A = \text{span} \{a'_1, \dots, a'_n\}$   
 $= \mathbb{F}^n$  by the fact  
 $Ax = b$  has solution

for all  $b$ ,

$\{a'_1, \dots, a'_n\}$  lin. ind.

thus the sol.  $x_0$  is unique

s.t. the only  $x$  satisfying  $Ax = 0$   
is  $x_0 = 0$   $\square$

Explain why this is the matrix form of the equivalence between injectivity and surjectivity for a linear operator on a finite-dimensional vector space.

Lin operator implies  $T(V, V)$

$T$  inj.  $\Leftrightarrow \ker T = \{0\}$  <sup>one to one</sup>

$T$  surj.  $\Leftrightarrow \text{im } T = V$  <sup>onto</sup>

if  $A_T x = 0$  has only 0 solution,  
only  $x$  mapped by  $A_T$  to 0  
is 0, such that  $\ker A_T = \{0\}$

if  $\forall b \in V, \exists x \in V$  s.t.  $A_T x = b$ ,

then entire codomain  $V = \text{span}\{b_1, \dots, b_n\}$   
is mapped by  $A_T$  to

s.t.  $\text{im } T = V \Rightarrow T$  surj.

8. Let  $P \in \mathcal{L}(V)$  satisfy  $P^2 = P$ . Prove that

$$V = \text{Ker } P \oplus \text{Im } P.$$

Calclit solve

Then, for an arbitrary  $v \in V$ , find explicit formulas for its component in  $\text{Ker } P$  and its component in  $\text{Im } P$ .

$$P: V \rightarrow V \quad P^2 = P \Rightarrow P_x = P(P_x)$$

$$\text{Ker } P = \{ v \in V : P v = 0 \}$$

$$\text{Im } P = \{ y = P x \mid \forall x \in V \}$$

1. **Fixing the Image:**  $P$  acts as the identity operator on its own range.

$$\exists u \in V \text{ s.t. } w = Pu \implies Pw = P(Pu) = P^2u = Pu = w$$

2. **Complementary Annihilation:** The identity operator decomposes as  $I = (I - P) + P$ , where:

$$P(I - P) = P - P^2 = P - P = 0$$

This guarantees that for any  $v \in V$ , the vector  $(I - P)v$  lies in  $\text{Ker } P$ .

8. Let  $P \in \mathcal{L}(V)$  satisfy  $P^2 = P$ . Prove that

$$V = \text{Ker } P \oplus \text{Im } P.$$

Calclit solve

Then, for an arbitrary  $v \in V$ , find explicit formulas for its component in  $\text{Ker } P$  and its component in  $\text{Im } P$ .

$$P: V \rightarrow V \quad P^2 = P \Rightarrow P_x = P P_x$$

$$\text{Ker } P = \{ v \in V : P v = 0 \}$$

$$\text{Im } P = \{ y = P x \mid \forall x \in V \}$$

$P$  acts like a projection, s.t.

$$\exists u \in V \text{ s.t. } P u = w$$

$$\Rightarrow P P u = P w$$

$$P u = P w$$

$$w = P w \Rightarrow \forall w \text{ in image of } P, \\ P \text{ acts as } I$$

Compl. Annihilation

$$\forall v \in V, (\bar{I} - P)v = 0$$

### 3. Formal Proof: $V = \text{Ker } P \oplus \text{Im } P$

To establish that  $V$  is the internal direct sum of  $\text{Ker } P$  and  $\text{Im } P$ , we must show:

1.  $\text{Ker } P \cap \text{Im } P = \{0\}$
2.  $V = \text{Ker } P + \text{Im } P$

Part 1: Trivial Intersection ( $\text{Ker } P \cap \text{Im } P = \{0\}$ )

Let  $v \in \text{Ker } P \cap \text{Im } P$ .

- Since  $v \in \text{Ker } P$ ,  $Pv = 0$ .
- Since  $v \in \text{Im } P$ , there exists  $u \in V$  such that  $v = Pu$ .

Applying  $P$  to  $v$ :

$$v = Pu = P^2u = P(Pu) = Pv$$

Since  $Pv = 0$ , it follows directly that:

$$v = 0$$

Hence,  $\text{Ker } P \cap \text{Im } P = \{0\}$ .

Claim:

$$V = \text{Ker } P \oplus \text{Im } P$$

To prove:

$$1) \text{Ker } P \cap \text{Im } P = \{0\}$$

$$2) V = \text{Ker } P + \text{Im } P$$

$$1) \text{ let } v \in \text{Ker } P \cap \text{Im } P$$

$$\text{ s.t. } v \in \text{Ker } P$$

$$\Rightarrow Pv = 0 \quad (*)$$

$$\text{ since } v \in \text{Im } P,$$

$$\exists u \in V \text{ s.t.}$$

$$v = Pu$$

$$\Rightarrow v = Pu = P^2u = P(Pu) = Pv$$

$$\text{ by } (*), \quad Pv = 0 \Rightarrow v = 0$$

$$\text{ thus } \text{Ker } P \cap \text{Im } P = \{0\}$$

## Part 2: Spanning the Space ( $V = \text{Ker } P + \text{Im } P$ )

Let  $v \in V$  be arbitrary. Decompose  $v$  algebraically:

$$v = (v - Pv) + Pv$$

We verify the subspace membership for each term:

- For  $Pv$ : By definition,  $Pv \in \text{Im } P$ .
- For  $v - Pv$ : Applying the linear operator  $P$ :

$$P(v - Pv) = Pv - P^2v = Pv - Pv = 0$$

Therefore,  $(v - Pv) \in \text{Ker } P$ .

Because every  $v \in V$  can be expressed as the sum of an element in  $\text{Ker } P$  and an element in  $\text{Im } P$ , we have  $V = \text{Ker } P + \text{Im } P$ .

Combining Parts 1 and 2 completes the proof:

$$V = \text{Ker } P \oplus \text{Im } P$$

$$2) \forall v \in V$$

$$V = v - Pv + Pv \quad (*)$$

$$\text{Since } Pv - Pv = 0$$

$$\text{Split into } v - Pv \text{ \& } Pv$$

$$i) \text{ by def } Pv \in \text{Im } P$$

$$ii) \text{ apply } P \text{ to } v - Pv$$

$$P(v - Pv) = Pv - P^2v$$

$$= Pv - Pv = 0$$

$$\Rightarrow P(v - Pv) = 0$$

$$\& (v - Pv) \in \text{Ker } P$$

Since  $\forall v \in V$ ,  $V$  is sum of elem of  $\text{Im } P$  &  $\text{Ker } P$ ,  $V = \text{Ker } P + \text{Im } P$

Thus by 1 & 2,  
 $V = \text{Ker } P \oplus \text{Im } P$



## IV. Eigenvalues and How to Diagonalize a Matrix

**Mini-lecture.** For a square matrix  $A$ , an eigenvalue  $\lambda$  is a scalar for which

$$(A - \lambda I)v = 0$$

has a nonzero solution. For a  $2 \times 2$  matrix, the computation can be organized by the characteristic equation

$$\det(A - \lambda I) = 0, \quad \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Once  $\lambda$  is known, find its eigenvectors by row-reducing  $A - \lambda I$ . If an  $n \times n$  matrix has  $n$  linearly independent eigenvectors  $v_1, \dots, v_n$ , place them in the columns of

$$X = (v_1 \ \cdots \ v_n)$$

and put the corresponding eigenvalues on the diagonal of  $D$ . Then

$$AX = XD, \quad \text{hence} \quad A = XDX^{-1}.$$

This is the computational meaning of diagonalization.

**Worked example.** Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

The characteristic equation is

$$\det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = (\lambda - 5)(\lambda - 2) = 0.$$

Thus the eigenvalues are 5 and 2.

For  $\lambda = 5$ ,

$$(A - 5I)v = 0 \implies v \in \text{span}\{(1, 1)^T\}.$$

For  $\lambda = 2$ ,

$$(A - 2I)v = 0 \implies v \in \text{span}\{(1, -2)^T\}.$$

Choose

$$X = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The columns of  $X$  are independent, so  $X$  is invertible, and  $AX = XD$ . Therefore

$$A = XDX^{-1}.$$

This is the pattern to imitate in the next computational problems.

9. First eigenvalue calculation. Define  $T \in \mathcal{L}(\mathbb{F}^2)$  by

$$T(w, z) = (z, w).$$

(a) Write the matrix of  $T$  in the standard basis.

(b) Find all eigenvalues.

(c) For each eigenvalue, solve the corresponding homogeneous system to find the eigenspace.

(d) Choose an eigenbasis and write the diagonal matrix of  $T$  in that basis.

c) swaps  $v_1$  &  $v_2$  of  $\vec{v}$ ,

$$M_T = A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix}$$

$$h) |A - \lambda I| = 0 \quad \left| \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \right| = 0$$

$$\lambda^2 - 1 = 0 \Rightarrow (\lambda + 1)(\lambda - 1)$$

$$\lambda_1 = 1, \quad \lambda_2 = -1$$

$$c) \quad (A - \lambda_1 I)x = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_2 = 0$$

$$x_1 - x_2 = 0$$

$$x_1 = x_2$$

$$2) (A - \lambda_2 I)x = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 = 0$$

$$x_1 + x_2 = 0$$

$$v_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

$$v_2 = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

d) Choose  $X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$   $\det X = -1 - 1 = -2 \neq 0$   
 $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$   $X$  is invert.

$$AX = XD$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

thus  $D = X^{-1}AX$

w/  $X^{-1} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$   
 $= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

10. A matrix that does not have enough eigenvectors. Define  $T \in \mathcal{L}(\mathbb{F}^3)$  by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3).$$

(a) Write the standard matrix  $A$  of  $T$ .

(b) Find all eigenvalues and their eigenspaces.

(c) Count the number of linearly independent eigenvectors that can be obtained.

(d) Decide whether  $A$  can be diagonalized, and explain the answer using your calculation rather than quoting a theorem without verification.

$$a) \begin{matrix} 3 \times 3 & 3 \times 1 & 3 \times 1 \\ \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} & \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} & = \begin{bmatrix} 2z_2 \\ 0 \\ 5z_3 \end{bmatrix} \\ A & \underline{z} & \underline{z}' \end{matrix}$$

$$b) \det(A - \lambda I) = 0$$

$$\begin{vmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & 5-\lambda \end{vmatrix} = 0$$

$$-\lambda \begin{vmatrix} -\lambda & 0 \\ 0 & 5-\lambda \end{vmatrix} - 2 \begin{vmatrix} 0 & 0 \\ 0 & -\lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & -\lambda \\ 0 & 0 \end{vmatrix} = 5\lambda^2 - \lambda^3 = 0$$

$$-\lambda(\lambda^2 - 5\lambda)$$

$$5\lambda^2 - \lambda^3$$

$$\lambda^2(\lambda - 5) = 0$$

$$\lambda_1 = 5, \lambda_2 = 0, \lambda_3 = 0$$

c) multiplicity of 2 for  $\lambda_2 = \lambda_3 = 0$   
so only 2 lin ind ev

$$d) (A - \lambda I)v = 0$$

$$\begin{bmatrix} -5 & 2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix} v = 0$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} v = 0$$

$$v_2 = 0 \Rightarrow v_1 = 0 \quad v_3 \text{ free}$$

$$\begin{bmatrix} 0 \\ 0 \\ v_3 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ 0 \\ 0 \end{bmatrix} \text{ for } x_2, x_3$$

multiplicity  $\Rightarrow$  X not lin ind  
of  $x_2, x_3$  X non-ind, A not diag.

# 11. Diagonalization and powers. Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

- Find the eigenvalues of  $A$ .
- Find one eigenvector for each eigenvalue.
- Form the eigenvector matrix  $X$  and the diagonal matrix  $D$ .
- Verify directly that  $AX = XD$ , and hence that  $A = XDX^{-1}$ .
- Use

$$A^k = XD^kX^{-1}$$

to derive a closed-form expression for  $A^k$ .

$$a) \det \begin{pmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{pmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - (2)(0) = 0$$

$$3 - 3\lambda - \lambda + \lambda^2 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 1)(\lambda - 3) = 0$$

$$\lambda_1 = 1 \quad \lambda_2 = 3$$

b)

$$\lambda_1 \begin{pmatrix} 1-1 & 2 \\ 0 & 3-1 \end{pmatrix} v = 0$$

$$\begin{pmatrix} 0 & 2 \\ 0 & -2 \end{pmatrix} v = 0$$

$$2v_2 = 0$$

$$-2v_2 = 0$$

$v_1$  free

$$v_2 = 0 \Rightarrow x = \begin{bmatrix} v_1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} v_1 = v_2 \Rightarrow x_2 = \begin{bmatrix} v_1 \\ 1 \end{bmatrix}$$

$$\lambda_2 \begin{pmatrix} 1-3 & 2 \\ 0 & 3-3 \end{pmatrix} v = 0$$

$$\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} v = 0$$

$$-2v_1 + 2v_2 = 0$$

$$v_1 = v_2 \Rightarrow x_1 = \begin{bmatrix} v_1 \\ v_1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} v_1 = x_2 \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$c) X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$d) AX = XD$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 3 \end{bmatrix} \quad \checkmark$$

$$\det X = 1, \quad X^{-1} \text{ exists}$$

$$e) A^k = X D^k X^{-1}$$

$$X^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$X X^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1^k & 3^k \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= 1^k + 3^k$$

$$\begin{bmatrix} 1^k & 2^k \\ 0 & 3^k \end{bmatrix} = A^k$$

12. Proof: similarity preserves eigenvalues. Let  $T \in \mathcal{L}(V)$  and let  $S \in \mathcal{L}(V)$  be invertible.

(a) Prove that  $T$  and  $S^{-1}TS$  have the same eigenvalues.

(b) If  $v$  is an eigenvector of  $T$  with eigenvalue  $\lambda$ , identify an eigenvector of  $S^{-1}TS$  with the same eigenvalue.

Explain why changing basis changes the matrix representing a linear operator but not its eigenvalues.

$$\begin{aligned} & \text{a) let } S^{-1}TS = B \\ & \exists v_T, v_B \in V \text{ eigenvectors w/ } v_T \neq 0, v_B \neq 0, \\ & \text{s.t. } Tv_T = \lambda_T v_T \text{ \& } Bv_B = \lambda_B v_B \\ & \text{Since } S \text{ invertible, } S \text{ surj. \& inj.} \\ & \text{s.t. } Su = v_T \text{ for some } u \in V \text{ (surj)} \\ & \Rightarrow u = S^{-1}v_T \\ & \text{Since } v_T \neq 0, u = S^{-1}v_T \neq 0 \text{ (inj.)} \\ & \text{apply } S^{-1}TS \text{ to } u: \\ & S^{-1}TSu = S^{-1}T(Su) = S^{-1}Tv_T \\ & = S^{-1}\lambda_T v_T = \lambda_T S^{-1}v_T = \lambda_T u \\ & \Rightarrow S^{-1}TSu = \lambda_T u \\ & Bu = \lambda_T u \Rightarrow \lambda_T \text{ is eigenvalue} \\ & \text{for } B = S^{-1}TS \end{aligned}$$

Direction 1: If  $\lambda$  is an eigenvalue of  $T$ , then  $\lambda$  is an eigenvalue of  $S^{-1}TS$

1. Let  $\lambda$  be an eigenvalue of  $T$ . By definition, there exists a non-zero vector  $v \in V$  ( $v \neq 0$ ) such that:

$$Tv = \lambda v$$

2. Since  $S$  is invertible,  $S$  is surjective, so there exists a vector  $u \in V$  such that  $Su = v$  (or equivalently, define  $u = S^{-1}v$ ).
3. Because  $S$  is injective and  $v \neq 0$ , it follows that:

$$u = S^{-1}v \neq 0$$

4. Apply the operator  $S^{-1}TS$  to the vector  $u$ :

$$(S^{-1}TS)u = S^{-1}T(Su) = S^{-1}(Tv)$$

5. Substitute  $Tv = \lambda v$ :

$$S^{-1}(Tv) = S^{-1}(\lambda v) = \lambda(S^{-1}v) = \lambda u$$

6. Therefore,  $(S^{-1}TS)u = \lambda u$  with  $u \neq 0$ , proving  $\lambda$  is an eigenvalue of  $S^{-1}TS$  with corresponding eigenvector  $u = S^{-1}v$ .



Direction 2: If  $\lambda$  is an eigenvalue of  $S^{-1}TS$ , then  $\lambda$  is an eigenvalue of  $T$

- Let  $(S^{-1}TS)w = \lambda w$  with  $w \neq 0$ .
- Left-multiply by  $S$ :

$$S(S^{-1}TS)w = S(\lambda w) \implies T(Sw) = \lambda(Sw)$$

- Setting  $z = Sw$ , since  $S$  is invertible and  $w \neq 0$ ,  $z \neq 0$ .
- Hence  $Tz = \lambda z$  with  $z \neq 0$ , proving  $\lambda$  is an eigenvalue of  $T$ .

$$\text{spec}(T) = \text{spec}(S^{-1}TS)$$

for  $\lambda_B$ ,  $Bv_B = \lambda_B v_B$

$$S^{-1}TSv_B = \lambda_B v_B \quad \text{for } v_B \neq 0$$

left multiply by  $S$ :

$$S(S^{-1}TS)v_B = S(\lambda_B v_B)$$

$$\text{let } Sv_B = z$$

$$TSv_B = \lambda_B Sv_B$$

$$\implies Tz = \lambda_B z, \quad \text{thus } \lambda_B \text{ is eigenvalue of } T$$

Since  $\lambda_B$  is w at  $T$  &  $\lambda_T$  is ev of  $B$ ,  
both have same evs

(b) If  $v$  is an eigenvector of  $T$  with eigenvalue  $\lambda$ , identify an eigenvector of  $S^{-1}TS$  with the same eigenvalue.

for  $v_T$  eigenvector of  $T$  w/  $\lambda_T$

$$Tv_T = \lambda_T v_T$$

find  $v_B$  for  $S^{-1}TS = B$  s.t.

$$Bv_B = \lambda_T v_B$$

by 4) all  $\lambda_T$  are eigenvalues  
of  $B$  s.t.  $\exists v_B$  associated  
s.t.  $Bv_B = \lambda_T v_B$  w/  $\lambda_T$

thus  $v_B$  shares  $\lambda_T$

w/  $v_T$

Explain why changing basis changes the matrix representing a linear operator but not its eigenvalues.

Matrix composed of vectors w/ coord.  
in given basis, changing basis  
adjusts coord but maintains  
inherent position vectors,  
so degree of scaling in the  
principal directions is invariant

# Just for studying purposes:

A linear operator  $T : V \rightarrow V$  is a coordinate-free geometric mapping on a vector space  $V$ . A matrix representation of  $T$  depends explicitly on the choice of basis used to parametrize  $V$ , whereas the eigenvalues of  $T$  are intrinsic algebraic and geometric properties of the operator itself.

## 1. Effect of a Change of Basis on the Matrix Representation

Let  $V$  be an  $n$ -dimensional vector space over a field  $\mathbb{F}$ , and let  $\mathcal{B} = \{v_1, \dots, v_n\}$  and  $\mathcal{B}' = \{v'_1, \dots, v'_n\}$  be two distinct ordered bases for  $V$ .

- The matrix representation of  $T$  with respect to  $\mathcal{B}$ , denoted  $[T]_{\mathcal{B}} = A \in \mathbb{F}^{n \times n}$ , is defined such that for any vector  $x \in V$ :

$$[T(x)]_{\mathcal{B}} = A[x]_{\mathcal{B}}$$

where the  $j$ -th column of  $A$  contains the coordinates of  $T(v_j)$  relative to  $\mathcal{B}$ .

- Let  $P = [I]_{\mathcal{B}}^{\mathcal{B}'} \in \text{GL}_n(\mathbb{F})$  be the invertible change-of-basis matrix converting  $\mathcal{B}'$ -coordinates to  $\mathcal{B}$ -coordinates:

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$$

- Expressing the action of  $T$  in the  $\mathcal{B}'$  basis yields:

$$[T(x)]_{\mathcal{B}'} = P^{-1}[T(x)]_{\mathcal{B}} = P^{-1}A[x]_{\mathcal{B}} = P^{-1}AP[x]_{\mathcal{B}'}$$

- Consequently, the matrix representation with respect to  $\mathcal{B}'$  is:

$$B = [T]_{\mathcal{B}'} = P^{-1}AP$$

Because  $P \neq I$  in general,  $B \neq A$ . A change of basis modifies the matrix representation via a similarity transformation.

- **Coordinate-Free Geometric Characterization:**

A scalar  $\lambda \in \mathbb{F}$  is an eigenvalue of  $T$  if and only if there exists a non-zero vector  $v \in V$  such that:

$$(T - \lambda I)v = 0$$

This definition relies solely on the action of  $T$  on the underlying vector space  $V$ . Since vectors and linear operators exist independently of coordinate frames, the existence of such a scalar  $\lambda$  and invariant subspace  $\ker(T - \lambda I)$  does not depend on the choice of basis.

- **Invariance of the Characteristic Polynomial:**

Let  $A = [T]_{\mathcal{B}}$  and  $B = P^{-1}AP = [T]_{\mathcal{B}'}$ . The characteristic polynomial  $p_B(\lambda)$  satisfies:

$$\det(\lambda I - B) = \det(\lambda I - P^{-1}AP) = \det(P^{-1}(\lambda I - A)P)$$

Applying the multiplicativity of the determinant:

$$p_B(\lambda) = \frac{1}{\det(P)} \det(\lambda I - A) \det(P) = \det(\lambda I - A) = p_A(\lambda)$$

Because  $A$  and  $B$  share identical characteristic polynomials, their roots—the eigenvalues  $\lambda_i$  along with their algebraic multiplicities—are strictly identical.

- **Explicit Eigenvector Coordinate Transformation:**

If  $x \in \mathbb{F}^n \setminus \{0\}$  is an eigenvector of  $A$  with eigenvalue  $\lambda$ , then  $Ax = \lambda x$ .

Transforming  $x$  into the coordinate system of  $\mathcal{B}'$  via  $y = P^{-1}x \neq 0$ :

$$By = (P^{-1}AP)(P^{-1}x) = P^{-1}Ax = P^{-1}(\lambda x) = \lambda(P^{-1}x)$$

Thus,  $y$  is an eigenvector of  $B$  corresponding to the identical eigenvalue  $\lambda$ , preserving both the eigenvalues and the geometric multiplicity  $\dim(\ker(A - \lambda I)) = \dim(\ker(B - \lambda I))$ .

## V. Orthogonality, Gram–Schmidt, and Projection

**Mini-lecture.** Given linearly independent vectors  $v_1, \dots, v_m$ , the Gram–Schmidt procedure constructs an orthonormal list spanning the same successive subspaces. For the first two vectors,

$$e_1 = \frac{v_1}{\|v_1\|}, \quad u_2 = v_2 - \langle v_2, e_1 \rangle e_1, \quad e_2 = \frac{u_2}{\|u_2\|}.$$

If  $e_1, \dots, e_m$  is an orthonormal basis of a subspace  $U$ , then the orthogonal projection of  $b$  onto  $U$  is

$$P_U b = \sum_{j=1}^m \langle b, e_j \rangle e_j.$$

The residual  $b - P_U b$  is orthogonal to every vector in  $U$ , which is why  $P_U b$  is the closest point in  $U$  to  $b$ .

**Worked example.** Apply Gram–Schmidt to  $v_1 = (1, 1)$  and  $v_2 = (1, 0)$  in  $\mathbb{R}^2$ :

$$e_1 = \frac{1}{\sqrt{2}}(1, 1),$$

$$u_2 = (1, 0) - \left\langle (1, 0), \frac{1}{\sqrt{2}}(1, 1) \right\rangle \frac{1}{\sqrt{2}}(1, 1) = \frac{1}{2}(1, -1),$$

so

$$e_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Thus  $e_1, e_2$  is an orthonormal basis spanning the same space as  $v_1, v_2$ .

13. On  $\mathcal{P}_2(\mathbb{R})$  define

$$\langle p, q \rangle = \int_0^1 p(x)q(x) dx.$$

Apply the Gram-Schmidt procedure to

$$1, x, x^2$$

to produce an orthonormal basis of  $\mathcal{P}_2(\mathbb{R})$ . Show every subtraction and normalization step.

$$V = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\langle v_1, v_1 \rangle}} = \frac{1}{\sqrt{\int_0^1 1 \cdot 1 dx}} = \frac{1}{\sqrt{1}} = 1$$

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1 = x - \langle x, 1 \rangle 1$$

$$= x - \int_0^1 x \cdot 1 dx = x - \left[ \frac{1}{2} x^2 \right]_0^1 = x - \frac{1}{2}$$

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{x - 1/2}{\sqrt{\langle x - 1/2, x - 1/2 \rangle}} = \frac{x - 1/2}{\sqrt{\int_0^1 (x - 1/2)(x - 1/2) dx}} = \frac{x - 1/2}{\sqrt{\int_0^1 x^2 - x + 1/4 dx}} = \frac{x - 1/2}{\sqrt{\left[ \frac{1}{3} x^3 - \frac{1}{2} x^2 + \frac{1}{4} x \right]_0^1}} = \frac{x - 1/2}{\sqrt{\left( \frac{1}{3} - \frac{1}{2} + \frac{1}{4} \right)}} = \frac{x - 1/2}{\sqrt{1/12}} = \frac{1}{\sqrt{3}} \left( x - \frac{1}{2} \right)$$

$$u_3 = v_3 - \langle v_3, e_2 \rangle e_2 - \langle v_3, e_1 \rangle e_1$$

$$x^2 - \left\langle x^2, \frac{x - 1/2}{\sqrt{1/12}} \right\rangle \frac{x - 1/2}{\sqrt{1/12}} - \langle x^2, 1 \rangle 1$$

$$x^2 - \int_0^1 x^2 \left( \frac{x - 1/2}{\sqrt{1/12}} \right) dx \left( \frac{x - 1/2}{\sqrt{1/12}} \right) - \int_0^1 x^2 dx$$

$$x^2 - \int_0^1 x^2 \left( \frac{x-1/2}{\sqrt{1/2}} \right) dx \left( \frac{x-1/2}{\sqrt{1/2}} \right) - \int_0^1 x^2 dx$$

$$x^2 - \frac{x-1/2}{1/2} \int_0^1 x^3 - \frac{1}{2} x^2 dx = \left[ \frac{1}{3} x^3 \right]_0^1$$

$$x^2 - 12(x-1/2) \left[ \frac{1}{4} x^4 - \frac{1}{2} x^3 \right]_0^1 = \left( \frac{1}{3} \right)$$

$$x^2 - (12x-6) \left( \frac{1}{4} - \frac{1}{6} \right) = \frac{1}{3} \quad \frac{3}{2} - 1 = \frac{1}{2}$$

$$x^2 - \left( 3x - 2x - \frac{3}{2} + 1 \right) = \frac{1}{3} = x^2 - x + \frac{1}{6} = u_3$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{x^2 - x + 1/6}{\sqrt{\langle x^2 - x + 1/6, x^2 - x + 1/6 \rangle}}$$

$$= \frac{x^2 - x + 1/6}{\sqrt{\int_0^1 (x^2 - x + 1/6)(x^2 - x + 1/6) dx}}$$

$$= \frac{x^2 - x + 1/6}{\sqrt{\int_0^1 x^4 - x^3 + \frac{1}{6} x^2 - x^3 + x^2 - \frac{1}{6} x + \frac{1}{6} x^2 - \frac{1}{6} x + \frac{1}{36} dx}}$$

$$\int_0^1 x^4 - x^3 + \frac{1}{6} x^2 - x^3 + x^2 - \frac{1}{6} x + \frac{1}{6} x^2 - \frac{1}{6} x + \frac{1}{36} dx$$

$$x^5 - 2x^3 + \frac{4}{3} x^2 - \frac{1}{3} x + \frac{1}{36}$$

$$\left[ \frac{1}{5} x^5 - \frac{1}{2} x^4 + \frac{4}{9} x^3 - \frac{1}{6} x^2 + \frac{1}{36} x \right]_0^1$$

$$\frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} =$$

$$= \frac{x^2 - x + 1/6}{\sqrt{1/180}} = 6\sqrt{5} \left( x^2 - x + \frac{1}{6} \right) = e_3$$

$$= \sqrt{5} (6x^2 - 6x + 1)$$

Orthogonal

Basis =  $\{e_1, e_2, e_3\} =$

$$\left\{ 1, \frac{1}{2\sqrt{3}} \left( x - \frac{1}{2} \right), \sqrt{5} (6x^2 - 6x + 1) \right\}$$



14. In  $\mathbb{R}^4$ , let

$$U = \text{span}\{(1, 1, 0, 0), (1, 1, 1, 2)\}.$$

Find the vector  $u \in U$  that minimizes

$$\|u - (1, 2, 3, 4)\|.$$

You may first orthonormalize a basis of  $U$ . After finding  $u$ , verify explicitly that the residual  $(1, 2, 3, 4) - u$  is orthogonal to both spanning vectors of  $U$ .

a) Given

$$e_1 = \frac{v_1}{\|v_1\|}, \quad u_2 = v_2 - \langle v_2, e_1 \rangle e_1, \quad e_2 = \frac{u_2}{\|u_2\|}.$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$e_1 = \frac{v_1}{\|v_1\|_2} = \frac{(1 \ 1 \ 0 \ 0)^T}{\sqrt{1^2 + 1^2 + 0^2 + 0^2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix}$$

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} - \left( \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0 + 0 \right) \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} - \frac{2}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 1-1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{u_2}{\sqrt{0+0+1+4}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$B_e = \left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \right\}$$

let  $v = (1, 2, 3, 4)$  in original basis  $U$

$$\min u = P_U(v) = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2$$

by projecting  $v$  w/  $B_e$

$$P_u(v) = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2$$

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} \quad e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\frac{3}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} + \frac{11}{\sqrt{5}} \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} 3/2 \\ 3/2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 11/5 \\ 22/5 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \\ 11/5 \\ 22/5 \end{bmatrix} = u$$

Verify

$$r = v - u = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 11/5 \\ 22/5 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 4/5 \\ -2/5 \end{bmatrix}$$

$$r \perp v_1 = (1, 1, 0, 0)^T ?$$

$$\langle r, v_1 \rangle = -1/2 + 1/2 + 0 + 0 = 0 \quad \checkmark$$

$$r \perp v_2 = (1, 1, 1, 2)^T ?$$

$$\langle r, v_2 \rangle = -\frac{1}{2} + \frac{1}{2} + \frac{4}{5} - \frac{4}{5} = 0 \quad \checkmark$$

$$u = (3/2, 3/2, 11/5, 22/5)^T$$

15. Proof: orthogonality as a minimization condition. Suppose  $u, v \in V$ . Prove that

$$\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\| \text{ for every } a \in \mathbb{F}.$$

Interpret the result as the simplest version of the least-squares first-order condition.

$$\Rightarrow \text{if } \langle u, v \rangle = 0,$$

let  $a \in \mathbb{F}$  be some scalar,

by inner prod prop  $\|u\|^2 = \langle u, u \rangle$

$$\text{we know } \|u + av\|^2 = \langle u + av, u + av \rangle \\ = \langle u, u + av \rangle + \langle av, u + av \rangle \\ \text{by lin in first arg}$$

$$= \langle u, u \rangle + a \langle u, v \rangle + \bar{a} \langle v, u \rangle + |a|^2 \langle v, v \rangle \\ \text{by conj- lin in 2nd}$$

thus

$$\|u + av\|^2 = \langle u, u \rangle + a \langle v, u \rangle + \bar{a} \langle v, u \rangle + |a|^2 \langle v, v \rangle \\ \langle u, u \rangle = \|u\|^2 \quad \& \quad \langle v, v \rangle = \|v\|^2$$

$$\|u + av\|^2 = \|u\|^2 + a \langle v, u \rangle + \bar{a} \langle v, u \rangle + |a|^2 \|v\|^2 \\ \text{since } \langle u, v \rangle = 0,$$

$$\|u + av\|^2 = \|u\|^2 + |a|^2 \|v\|^2$$

$$\text{since } |a|^2 \|v\|^2 \geq 0,$$

$$\|u + av\|^2 \geq \|u\|^2$$

sqrt both sides

$$\|u + av\| \geq \|u\| \quad \& \quad \forall a \in \mathbb{F}$$

⇐ if  $\|u\| < \|u + \alpha v\| \quad \forall \alpha \in \mathbb{F}$ ,  
symm both sides

$$\|u\|^2 \leq \|u + \alpha v\|^2$$

$$\|u\|^2 \leq \|u\|^2 + 2\bar{\alpha} \langle u, v \rangle + |\alpha|^2 \|v\|^2$$

$$\Rightarrow 0 \leq 2\bar{\alpha} \langle u, v \rangle + |\alpha|^2 \|v\|^2$$

if  $v = 0$ ,  $\langle u, v \rangle = 0$   $\forall \alpha$   $u_1(0) + \dots + u_n(0) = 0$

for  $v \neq 0$ ,  $0 \leq |\alpha|^2 \|v\|^2$

$$\Rightarrow 0 \geq -|\alpha| \|v\|^2$$

$$-|\alpha|^2 \|v\|^2 \leq 2\bar{\alpha} \langle u, v \rangle$$

$$-|\alpha|^2 \|v\|^2 < 0 \leq 2\bar{\alpha} \langle u, v \rangle$$

???

$$2\bar{\alpha} \langle u, v \rangle \leq 0 \leq 2\bar{\alpha} \langle u, v \rangle$$

thus  $\langle u, v \rangle = 0$

□

## 2. Non-Trivial Case ( $v \neq 0$ ):

Assume  $v \neq 0$ , so  $\|v\|^2 > 0$ . Expanding the squared norm for any  $a \in \mathbb{F}$ :

$$\|u + av\|^2 = \langle u + av, u + av \rangle = \|u\|^2 + a\langle v, u \rangle + \bar{a}\langle u, v \rangle + |a|^2\|v\|^2$$

Using  $\langle v, u \rangle = \overline{\langle u, v \rangle}$ , this becomes:

$$\|u + av\|^2 = \|u\|^2 + 2 \operatorname{Re}(\bar{a}\langle u, v \rangle) + |a|^2\|v\|^2$$

## 3. The Missing Step (Choosing $a$ ):

Since the inequality  $\|u\| \leq \|u + av\|$  holds for *all*  $a \in \mathbb{F}$ , choose the specific scalar:

$$a = -\frac{\langle u, v \rangle}{\|v\|^2}$$

had to look  
up solution

## 4. Substitution & Simplification:

Substitute this choice of  $a$  into the expansion:

$$\|u + av\|^2 = \|u\|^2 - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, u \rangle - \frac{\overline{\langle u, v \rangle}}{\|v\|^2} \langle u, v \rangle + \frac{|\langle u, v \rangle|^2}{\|v\|^4} \|v\|^2$$

$$\|u + av\|^2 = \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} - \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2}$$

$$\|u + av\|^2 = \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}$$

## 5. Conclusion:

The hypothesis  $\|u\|^2 \leq \|u + av\|^2$  yields:

$$\|u\|^2 \leq \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} \implies \frac{|\langle u, v \rangle|^2}{\|v\|^2} \leq 0$$

Because  $\|v\|^2 > 0$  and  $|\langle u, v \rangle|^2 \geq 0$ , it must be that:

$$|\langle u, v \rangle|^2 = 0 \implies \langle u, v \rangle = 0$$

## Least-Squares First-Order Condition Interpretation

In the optimization problem:

$$\min_{a \in \mathbb{F}} \|u + av\|^2$$

the objective function  $f(a) = \|u\|^2 + 2 \operatorname{Re}(\bar{a}\langle u, v \rangle) + |a|^2 \|v\|^2$  is strictly convex with respect to  $a$ . The point  $a = 0$  is the global minimizer if and only if the directional derivative (gradient) at 0 vanishes:

$$\nabla f(0) = 2\langle u, v \rangle = 0 \iff \langle u, v \rangle = 0$$

This expresses the classic projection theorem: the error vector  $u$  has minimal norm if and only if it is orthogonal to the subspace spanned by  $v$ .

## Optional Exploratory Problem

16. What failure of diagonalization looks like. Let  $T \in \mathcal{L}(\mathbb{C}^2)$  be defined by

$$T(w, z) = (z, 0).$$

- (a) Find all eigenvalues and eigenspaces of  $T$ .
- (b) Is there a basis of  $\mathbb{C}^2$  consisting of eigenvectors? Explain.
- (c) Compute  $T^2$ .
- (d) Find  $v$  such that  $Tv \neq 0$  but  $T^2v = 0$ .
- (e) Show that  $(Tv, v)$  is a basis and write the matrix of  $T$  in this basis.
- (f) The resulting matrix is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Explain informally what extra information this matrix records that ordinary eigenvectors miss.

$$\begin{aligned} \text{h)} \quad T(w, z) &= (z, 0) = A \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ 0 \end{bmatrix} \\ &\Rightarrow A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{sel} \quad \det(A - \lambda I) &= 0 \Rightarrow \det \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \end{bmatrix} = 0 \\ \lambda^2 - 0 &= 0 \Rightarrow \lambda^2 = 0 \\ \lambda_1 &= \lambda_2 = 0 \end{aligned}$$

$$Av = \lambda v = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} v_1 \text{ free} \\ v_2 = 0 \end{matrix} \Rightarrow v = \begin{bmatrix} t \\ 0 \end{bmatrix}$$

$$\text{eigenspace} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

b)  $N_2$  since only eigenvectors  
are of form  $\begin{bmatrix} t \\ 0 \end{bmatrix}$ ,  
they can only span 1  
dimension, so cannot  
form a basis for  $\mathbb{C}^2$

$$c) T^2 = T(T(w, z)) = T(z, 0) = (0, 0)$$

$$AA\begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} w \\ z \end{bmatrix} \\ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(d) Find  $v$  such that  $Tv \neq 0$  but  $T^2v = 0$ .

(e) Show that  $(Tv, v)$  is a basis and write the matrix of  $T$  in this basis.

d)  $T(v_1, v_2) \neq 0$   $T^2(v_1, v_2) = 0$   
applies for any  $v_2 \neq 0$

e)  $(Tv, v)$  for  $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$  w/  $v_2 \neq 0$   
 $= \left\{ \begin{bmatrix} v_2 \\ 0 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\}$  lin ind & spans  $\mathbb{C}^2$

$\forall c_1, c_2 \in \mathbb{C}, a_1 v_2 + a_2 v_1 = c_1$   
 $a_2 v_2 = c_2$   
 $\exists a_1, a_2 \Rightarrow$  by choice of  $\mathbb{C}$

since  
 $a_1 \begin{bmatrix} v_2 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$   
 $= \begin{bmatrix} a_1 v_2 \\ 0 \end{bmatrix} + \begin{bmatrix} a_2 v_1 \\ a_2 v_2 \end{bmatrix}$   
 $= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$   
only if  
 $a_2 v_2 = 0$   
 $\Rightarrow a_2 = 0$   
&  $a_1 v_2 = 0$   
 $\Rightarrow a_1 = 0$



e) Matrix at  $\tau$  in  $\left\{ \begin{bmatrix} v_2 \\ 0 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\}$   
 s.t.  $T(u, z) = (z, 0)$

$$= \left[ 0 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, 1 \begin{bmatrix} v_2 \\ 0 \end{bmatrix} \right]$$

$$\text{For } v_2 = 1$$

$v_1$  free

$$= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

The resulting matrix is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Explain informally what extra information this matrix records that ordinary eigenvectors miss.

The matrix  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  encodes **transient, step-by-step dynamical coupling** (a Jordan chain), which pure eigenvectors completely miss because eigenvectors only detect static directions that map strictly into themselves.

## The Information Missing from Ordinary Eigenvectors

- **Static Invariance vs. Directional Flow:**

- An ordinary eigenvector  $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  satisfies  $Tv_1 = 0 \cdot v_1 = 0$ . It only identifies a static 1D subspace (the kernel) where vectors are crushed to zero immediately.
- The second basis vector  $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$  is a **generalized eigenvector**. It satisfies  $(T - 0I)^2 v_2 = 0$ , but  $(T - 0I)v_2 \neq 0$ .
- Instead of mapping to a scalar multiple of itself,  $v_2$  maps directly to  $v_1$ :

$$Tv_2 = v_1, \quad Tv_1 = 0$$

- **Nilpotent Shift Structure:**

The off-diagonal 1 records a **chain of progression** (or shift operator):

$$v_2 \xrightarrow{T} v_1 \xrightarrow{T} 0$$

Ordinary eigenvectors only tell you the "terminal state" ( $v_1 \rightarrow 0$ ). The matrix entry 1 tells you that there is a 1-step "delay" or "queue" before reaching the kernel.

- **Dynamical and Differential Implications:**

In continuous dynamical systems ( $\dot{x} = Ax$ ), a diagonal matrix produces pure exponential decay or growth ( $e^{\lambda t}$ ). A non-zero superdiagonal 1 in a Jordan block produces secular polynomial modulation:

## Optional Exploratory Problem

**16. What failure of diagonalization looks like.** Let  $T \in \mathcal{L}(\mathbb{C}^2)$  be defined by

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